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# High Performance Computing

## Cluster

**Dr Ligang He**

# What you will learn today

- Fundamentals of cluster systems
- Cluster manager
- Applications and benchmarks for clusters (supercomputers)

# What is Cluster?

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- **A system built by connecting stand-alone computers with high-speed network (cable networks)**
  - Stand-alone computers
  - Commodity off the shelve computers
  - high-speed network: infiniband, Gigabit Ethernet, Intel Omni-path
  - Single system image: separate computers are managed and used as one unified system

# Why do Clusters gain popularity?

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- **Clustering gain interests when 3 technologies converged:**

  1. Very high performance Microprocessors

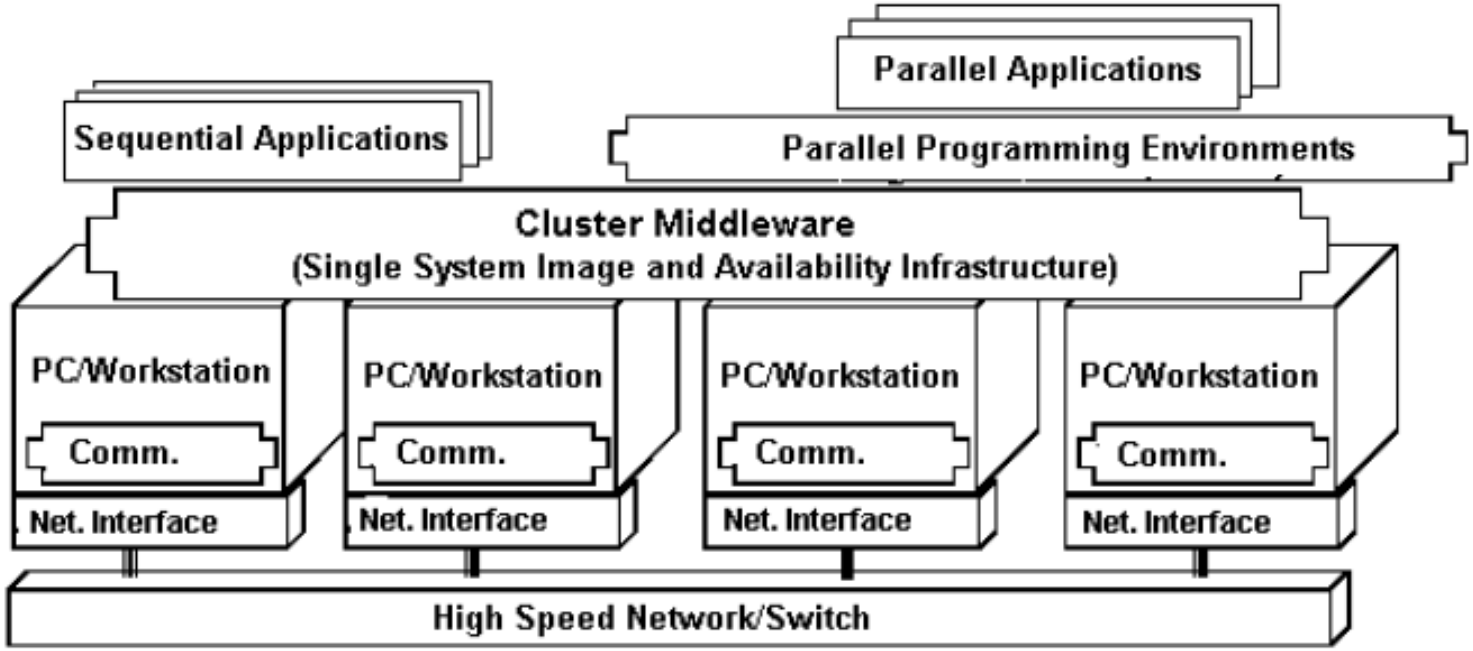
     - PC performance today = supercomputers at old times

  2. High speed networking

  3. Standard tools for parallel/ distributed programming

- **Advantage over custom-designed mainframe computers: Good portability and scalability**

# Cluster Architecture




Cluster system architecture

# What you will learn today

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- **Cluster manager**
- Applications and benchmarks for clusters (supercomputers)

# Cluster Manager

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- Help the allocation of resources to jobs, subject to jobs' resource requirements and policy restrictions in cluster
- Three parties in a cluster
  - Users: submit the job and job requirements
  - Administrators: describing policy restrictions in clusters
  - Cluster manager: monitoring the state of the cluster, scheduling the jobs and tracking the resource usage
- Typically, five activities are performed
  - Queuing
  - Scheduling
  - Monitoring
  - Resource management
  - Accounting

# Queuing

- Job submission usually consists of two primary parts:
  - Job description (e.g. job name, the location of the required input files)
  - Resource requirements (e.g. the number of CPUs, the amount of memory)

```
#!/bin/sh
#SBATCH -p cs402 # Queue
#SBATCH -N 2 # Number of nodes
#SBATCH --tasks-per-node 2 # Number of tasks per node
#SBATCH -t 00:05:00 # Time limit hrs:min:sec
srun ./helloWorldMPI
```

- Once submitted, the jobs are held in the queue until the matching resources are available

```
squeue -u your_username
```

# Scheduling

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- Determining at what time a job should be put into execution on which resources
- There are a variety of metrics to measure scheduling performance
  - System-oriented metrics (e.g. throughput, utilisation, average response time of all jobs)
  - user-oriented metrics (e.g. response time of a job submitted by a user)

# Monitoring

→ **Providing information to administrators, users and the Cluster manager on the status of jobs and resources**

```
squeue -u your_username
```

# Resource management

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→ **Handling the practical details of job execution and resource control**

- ❑ Allocate or release resources
- ❑ Start/stop/cancel jobs  scancel 271276
- ❑ Clean up after a job finishes (deleting temporary files generated during execution
- ❑ Removing or adding resources

→ **Job execution is offered in two modes**

- ❑ For the batch mode, the jobs are put into execution in such a way that the users don't have to be present during execution
- ❑ For interactive mode, the users have to be present to supply information during the execution.

# Accounting

→ Accounting for which users are using what resources for how long

# Schedule Polices

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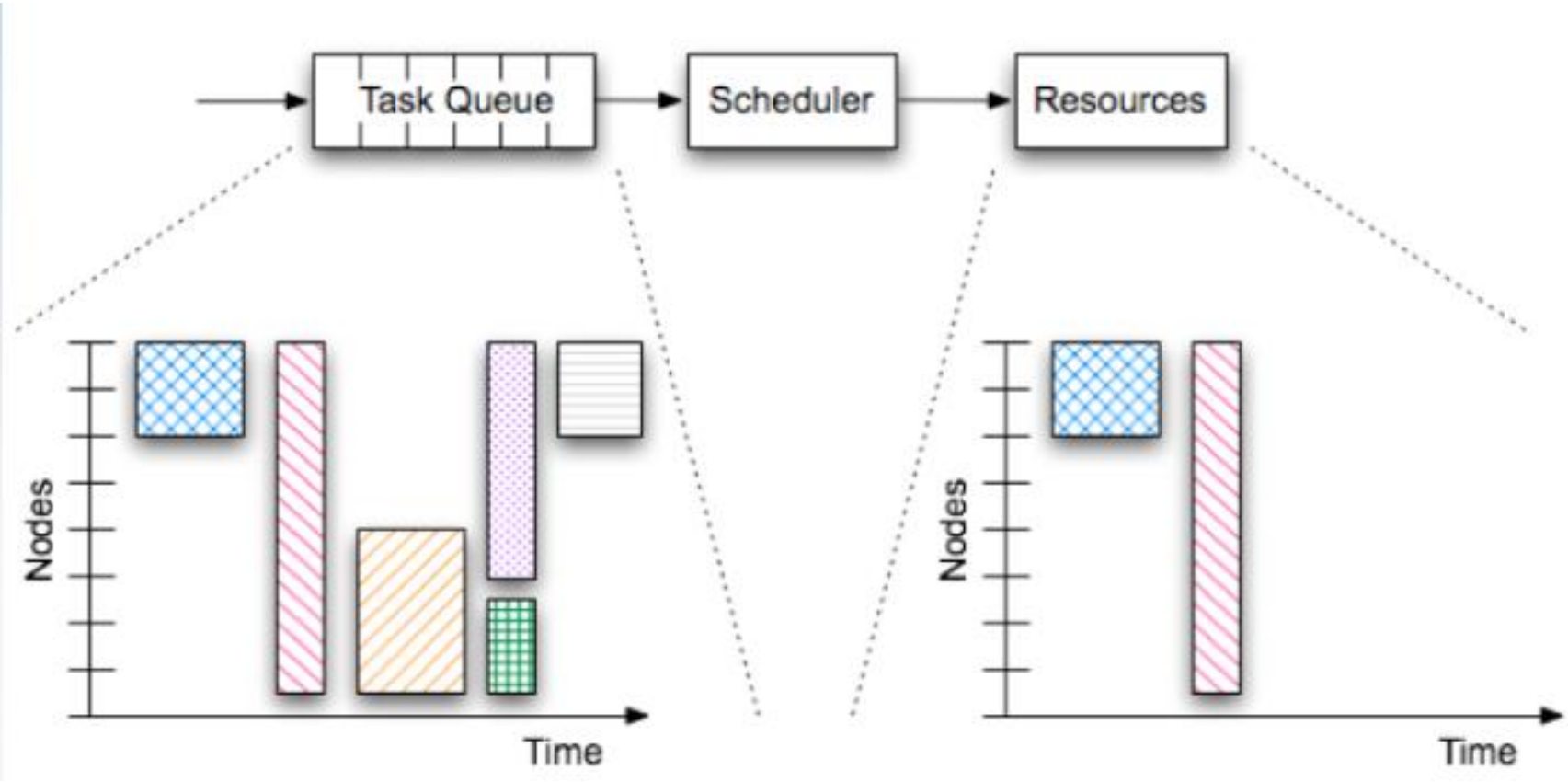
**The simplest policy:**

- **First-Come First-Served**
- **Jobs are run in the same order as they are submitted.**
- **Does not require prior knowledge about jobs (e.g. runtime).**
- **Problems: low resource utilization**
  - **jobs can block other jobs from starting, despite there being no benefit to either job.**

# First-Come First-Served



# Backfilling

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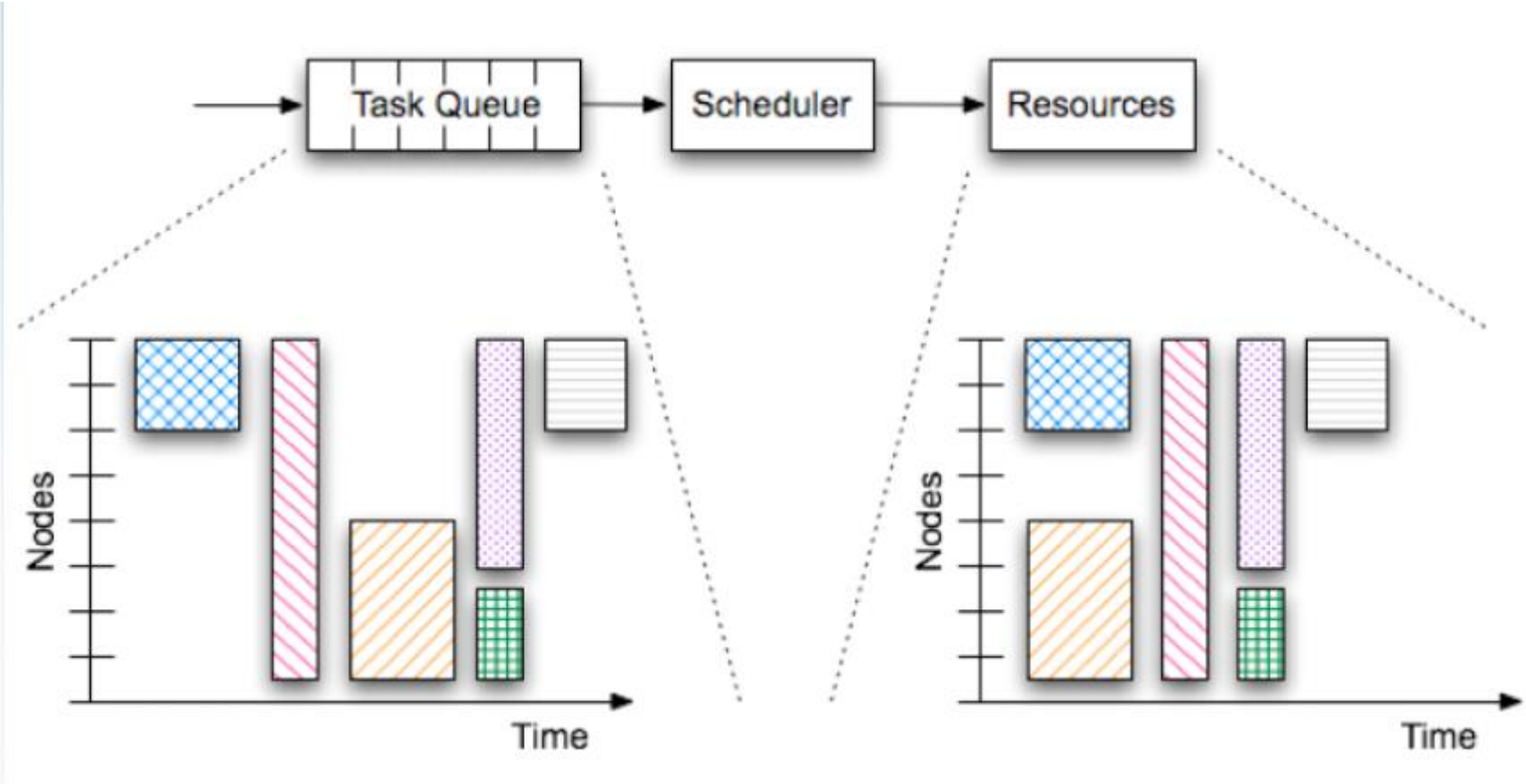
- The problem with FCFS is that idle time (sum of unused processing intervals) can be significant.
- One improvement is to “backfill”.
- Allows a job to start if it does not delay the execution of the first job in the queue.

# Backfilling



# Backfilling

- **Advantages:**
  - Utilisation is improved.
- **Disadvantages:**
  - Need to estimate the job execution time.
  - estimations are usually inaccurate.
  - It is a policy decision to decide what to do if a job overruns;
  - The default decision is to terminate the job if it overruns

# Backfilling

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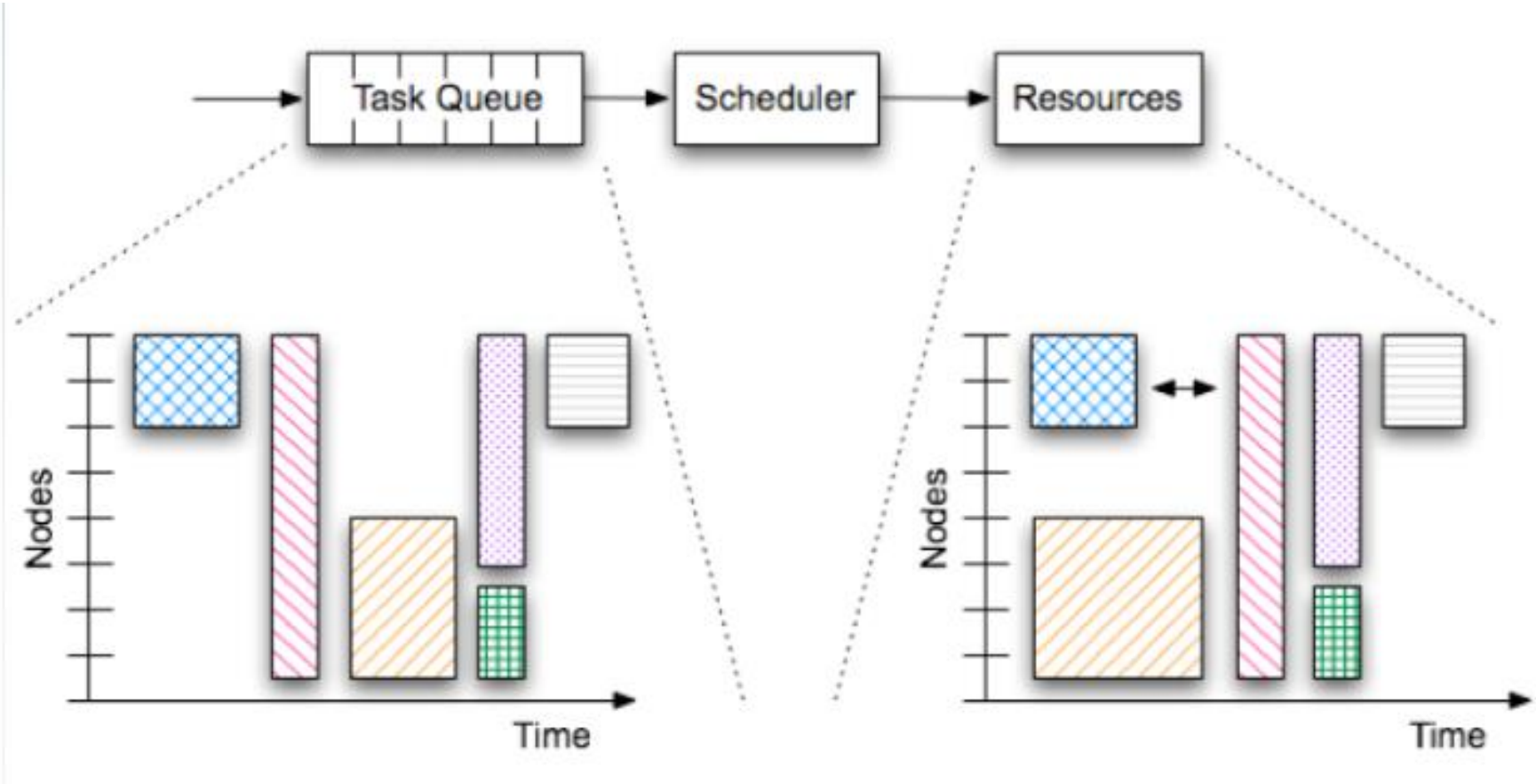
## A problem if predicted runtime is wrong:



# Priority-based Scheduling

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- Slurm uses a priority-based scheduling
- Priority of a job is calculated using a list of parameters, such as
  - Job requested resource
  - User's fair share
  - Job requested run time
  - Job age

# What you will learn today

- Fundamentals of cluster systems
- Cluster manager
- **Applications and benchmarks for clusters (supercomputers)**

# The Features of Scientific Applications

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- Many scientific applications solve linear equations
- The features of scientific applications
  - Calculating real numbers
  - Data to be processed are often stored using matrix data structure

# Solving Linear Equation Set

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→Repeating iterative steps and each step generates a better approximation of the solution

- Step 1: Guess a initial solution $\Phi^0$
- Step 2: Check if convergence is reached by checking the residual $b-A\Phi^i<\text{tolerance}$
- Step 3: For $\Phi^i$ $(i>=0)$, $\Phi^{(i+1)}= \Phi^{(i)}+(b-A\Phi^{(i)})$, go to Step 2

# The Features of Scientific Applications

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- Many scientific applications solve linear equations

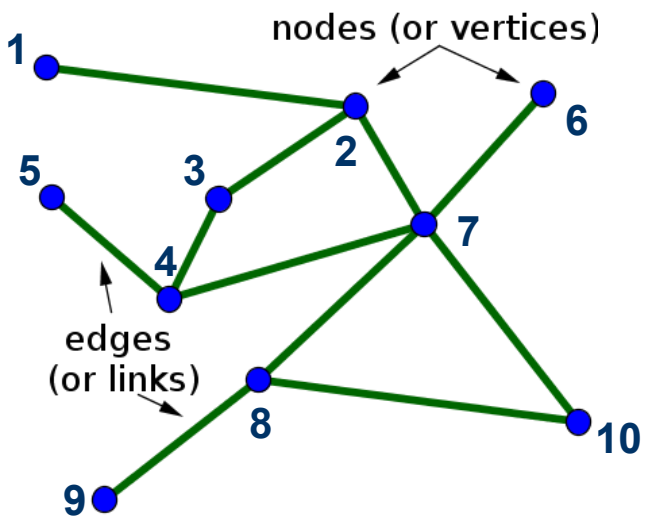
- The features of scientific applications
  - Calculating real numbers
  - Data to be processed are often stored using matrix data structure
  - High data locality (spatial locality and temporal locality)
  - Cache is critical in performance
    - Cache line (64 bytes)

- FLOPS is used to measure the ability of a computer system in processing real numbers
  - High Performance Linpack (HPL) benchmark is used to rank the supercomputers in Top500 list
  - HPL solves a dense system of linear equations

# New application areas

- In some applications, data are best stored with a graph structure
  - If using matrix, much space is wasted.
  - A graph structure consists of a set of ordered pairs (edges) of nodes

(1, 2), (2, 3), (2, 7), (3, 4), (4, 5), (4, 7), (6, 7), (7, 8), (7, 10), (8, 9), (8, 10)



# New application areas

- In some applications, data are best stored with a graph structure

- Examples of such new application areas
  - Studying human interaction patterns in social networking
  - Studying neuron interaction patterns in neuroscience
  - Online-shopping recommendation

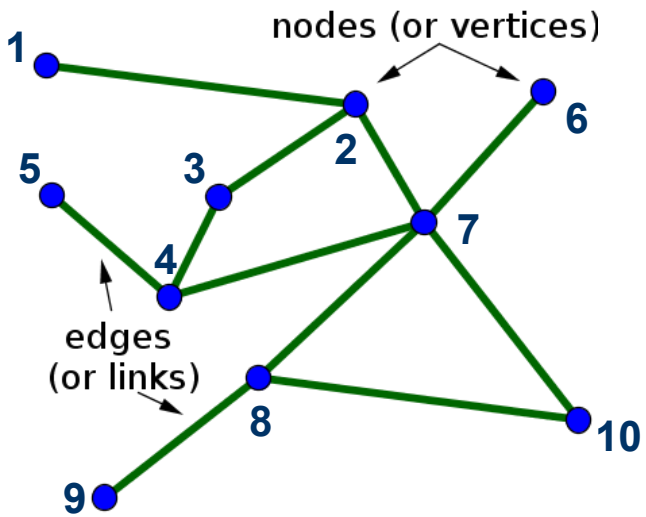
- Social networking

Neuroscience

# Graph-based applications show different features

- The data are best stored using graph data structure
- Have to store and access a large amount of data (intensive memory access)
- Need to establish the connections among data points (make the order of the edges more random)
- Data locality is low

(1, 2), (2, 3), (2, 7), (3, 4), (4, 5), (4, 7), (6, 7), (7, 8), (7, 10), (8, 9), (8, 10)



# Features of graph-based applications

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- **These applications show different features**
  - **Have to store and access a large amount of data (intensive memory access)**
  - **The data are best stored using graph data structure**
  - **Data locality is low**
  - **Cache now has limited positive impact on performance**

**Conclusion: A computer with a high FLOPS performance may not perform well with these graph-based applications**

# Challenging the Top 500 list

**GIGATEPS AHEAD:** The Intrepid supercomputer is at the top of the Graph 500 list.
*PHOTO: ARGONNE NATIONAL LABORATORY*

## Better Benchmarking for Supercomputers

The usual yardstick is not a good metric

So Murphy and his colleagues from other U.S. national laboratories, academia, and industry have put together a benchmark they're calling the Graph 500. The name comes from the field of mathematics (graph theory) that the benchmark draws most heavily from. And the 500 is, Murphy says, an "aspirational" figure representing what they hope someday will be a "top 500" ratings list of the highest-performing supercomputers around the world, measured in gigateps instead of gigaflops.

The current biannual Top 500 supercomputers list recently made headlines

# Graph 500

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- Measure how fast a computer can traverse the edges of a graph
- Metrics: "gigateps"
  - Means “billions of traversed edges per second”
  - A node in the graph represents a data point
  - An edge is a connection between two data points.
  - Represent how fast a computer system accesses its global memory
  - Use the breadth-first search as the benchmark
- Using gigateps, supercomputers have been slowing down
  - “Over the past 15 years, each 1,000-fold increase in flops has resulted in a 10-fold decrease in accessible memory. ” - Prof. Peter Kogge, Notre Dame University
- http://www.graph500.org/